Business Statistics

Employee Satisfaction and Statistical Analysis in Organizations

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1. Introduction:

Employee satisfaction, or job satisfaction, is the degree of fulfillment and happiness an individual experiences within employee's work context. It comprises various elements, such as interactions with management, job duties, rewards, and alignment with organizational values. A high level of employee satisfaction can lead to enhanced productivity, improved business outcomes, and a positive organizational image (Pistilli, 2023c).

Employee satisfaction surveys are frequently used by employers to evaluate their employees' contentment with their work experience, environment, and company culture. This survey includes carefully crafted questions aimed at identifying factors contributing to satisfaction or dissatisfaction among employees. Managers can then examine the results to pinpoint key influences on staff morale and take appropriate actions to resolve any issues. Adopting such a proactive approach can lead to enhanced staff retention, engagement, and overall organizational effectiveness (Qualaroo Editorial Team, 2024).

2. Literature Review about Employee Satisfaction using 5 peer-reviewed studies published in the last five years:

2.1. The Influence of Organizational Support on Employee Satisfaction During Times of Crisis:

Employee engagement encompasses two primary dimensions, first - vigor, exemplified by high energy and resilience, a willingness to exert effort, and perseverance through adversity; and second - dedication, involving deep immersion in work and experiencing feelings of significance, enthusiasm, inspiration, pride, and challenge. An additional aspect of engagement is absorption, characterized by heightened focus and enjoyment in work, resulting in altered time perception and difficulty detaching from the task (Nguyen & Trần, 2021b).

The capacity to foster employee engagement is integral to an organization's growth, with a direct link to heightened productivity. During periods of adversity like the COVID-19 pandemic, prioritizing employee engagement becomes increasingly essential for ensuring optimal organizational performance and efficiency (Alshaabani et al., 2021).

2.2. The Impact of Diversity and Inclusion Initiatives on Employee Satisfaction:

Boosting workplace diversity and inclusion contributes to increased employee engagement and happiness by establishing a positive work atmosphere. This creates a psychologically safe setting where everyone feels appreciated and encouraged to excel in their positions (Tenney et al., 2023). Inclusivity and diversity play a substantial role in employee retention, with workers more inclined to stay when feeling appreciated and experiencing a sense of belonging. A McKinsey study highlighting that companies with diverse leadership teams are 33% more likely to surpass competitors in profitability reinforces that organizations emphasizing inclusion and diversity are better positioned to secure and maintain high-quality talent (Loeb, 2023).

2.3. Exploring the Influence of Work-Life Balance Policies on Employee Satisfaction:

Balancing work and personal life facilitates effective integration between these realms, allowing individuals to engage in leisure activities, pursue personal growth opportunities, and fulfill familial duties without sacrificing their professional pursuits. This harmony promotes happiness in both areas, resulting in a more rewarding and fulfilling life. Emphasizing work-life balance yields positive effects on physical health as well. Those who prioritize personal time tend to adopt healthier habits such as balanced diets, sufficient sleep, and consistent exercise. These behaviors enhance overall health, decreasing the likelihood of chronic health conditions and work absences (*The Impact of Work-Life Balance on Employee Well-Being and Productivity*, 2024).

2.4. The Role of Organizational Justice in Fostering Employee Satisfaction and Trust:

Despite its relatively new emergence, the social exchange view on organizational justice contrasts the established value-based approach. Considering justice's philosophical nature, its implications and violations are more discernible in relation to intrinsic motivation rather than extrinsic factors. Value discrepancies arising from justice violations hinder intrinsic motivation. However, the value-based perspective has received limited focus within empirical research, predominantly owing to the dominant social exchange viewpoint and insufficient attention to intrinsic motivation's mediating role (Lee & Rhee, 2023).

2.5. The Influence of Job Insecurity on Employee Satisfaction and Performance: A Longitudinal Study:

Numerous studies have established that job insecurity greatly increases job stress for employees. As a major stressor, job instability causes heightened feelings of threat and loss by reducing employment predictability and controllability. Job insecurity entails considerable uncertainty regarding job security within an organization. This uncertainty restricts employees' capacity to predict or control job retention, leading them to doubt their ability to secure vital resources for their well-being, like social identity, income, social networks, and social status. Job insecurity also directly impedes the satisfaction of basic psychological needs, such as relatedness, competence, and autonomy (Kim & Kim, 2020).

3. Hypothesis Tests:

3.1. Frequencies:

Statistics

		Job Involvement	Job Satisfaction	Performance Rating
N	Valid	412	412	412
	Missing	0	0	0

Figure 1: Frequency Statistics

With 412 comprehensive responses across several categories, including Job Involvement, Job Satisfaction and Performance Rating the table provides a reliable foundation for frequency analysis. The absence of missing data ensures a solid dataset for examining patterns and trends within these key workplace dimensions.

Job Involvement

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Low	28	6.8	6.8	6.8
	Medium	102	24.8	24.8	31.6
	High	243	59.0	59.0	90.5
	Very High	39	9.5	9.5	100.0
	Total	412	100.0	100.0	

Figure 2: Frequency of Job Involvement

The "Job Involvement" table classifies job involvement into four categories: Low, Medium, High, and Very High, featuring corresponding figures for Frequency, Percent, Valid Percent, and Cumulative Percent. The data indicates that 59.0% (243 individuals) experienced high involvement, 24.8% (102 individuals) reported medium involvement, 9.5% (39 individuals) conveyed very high involvement, and 6.8% (28 individuals) indicated low involvement. Cumulative percentages display a gradual increase in job involvement levels, totalling 100.0% across the 412 responses.

Job Satisfaction

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Low	84	20.4	20.4	20.4
	Medium	85	20.6	20.6	41.0
	High	124	30.1	30.1	71.1
	Very High	119	28.9	28.9	100.0
	Total	412	100.0	100.0	

Figure 3: Frequency of Job Satisfaction

The "Job Satisfaction" table presents Frequency, Percent, Valid Percent, and Cumulative Percent, dividing job satisfaction into four categories: Low, Medium, High, and Very High. According to the data, 20.4% (84 individuals) reported low satisfaction, 20.6% (85 individuals) indicated medium satisfaction, 30.1% (124 individuals) expressed high satisfaction, and 28.9% (119

individuals) conveyed very high satisfaction. Cumulative percentages across the 412 responses display a gradual increase in job satisfaction levels, totalling 100.0%.

Performance Rating

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Excellent	345	83.7	83.7	83.7
	Outstanding	67	16.3	16.3	100.0
	Total	412	100.0	100.0	

Figure 4: Frequency of Performance Rating

In the "Performance Rating" table, performance is classified into two categories—Excellent and Outstanding—alongside values for Frequency, Percent, Valid Percent, and Cumulative Percent. The data shows that 16.3% (67 individuals) were assessed as having outstanding performance, whereas 83.7% (345 individuals) were rated as having excellent performance. Cumulative percentages across the 412 responses reach 100.0%, signifying all responses are accounted for. This data sheds light on the prevalence of exceptional performance among the surveyed individuals and provides an understanding of performance rating dispersion within the studied group.

3.2. Descriptive Statistics:

Descriptive Statistics

	N	Range	Minimum	Maximum	Mean	Std. Deviation	Variance	Skewness		Kurtosis	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error	Statistic	Std. Error
Education	412	4	1	5	3.00	1.001	1.002	-.351	.120	-.418	.240
Environment Satisfaction	412	3	1	4	2.75	1.094	1.198	-.321	.120	-1.214	.240
Job Involvement	412	3	1	4	2.71	.729	.532	-.552	.120	.254	.240
Job Satisfaction	412	3	1	4	2.67	1.099	1.208	-.257	.120	-1.250	.240
Performance Rating	412	1	3	4	3.16	.369	.137	1.835	.120	1.375	.240
Relationship Satisfaction	412	3	1	4	2.67	1.056	1.115	-.254	.120	-1.143	.240
Work Life Balance	412	3	1	4	2.75	.705	.497	-.619	.120	.526	.240
Valid N (listwise)	412										

Figure 5: Descriptive Statistics

The spectrum of education levels, ranging from 1 to 4, exhibits a mean of 3.00, revealing that the majority of individuals in the dataset hold a bachelor's degree. With a variance of 1.002 and a

standard deviation of 1.001, it is possible to assess the distribution of data points around the mean and evaluate the dataset's overall variability.

The range of 1 to 4 for Environment Satisfaction denotes varying levels of individual satisfaction, with a mean of 2.75 signifying a moderate degree of overall contentment. The variance of 1.198 and standard deviation of 1.094 measure the dispersion of satisfaction scores from the mean, shedding light on the diverse perspectives individuals hold regarding their environment.

3.3. Chi- Square Test:

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Gender * Job Involvement	412	100.0%	0	0.0%	412	100.0%

Figure 6: Chi- Square Test (Case Processing Summary)

The Case Processing Summary scrutinizes the association between Gender and Job Involvement, covering 412 valid instances from the dataset. Remarkably, there are no missing instances, leading to 0 counts and percentages, emphasizing the dataset's wholeness and lack of gaps. The precise classification of Gender and Job Involvement uncovers 412 legitimate instances, accounting for 100% of the interactions.

Gender * Job Involvement Crosstabulation

			Job Involvement				
			Low	Medium	High	Very High	Total
Gender	Female	Count	11	44	86	9	150
		Expected Count	10.2	37.1	88.5	14.2	150.0
		% within Job Involvement	39.3%	43.1%	35.4%	23.1%	36.4%
	Male	Count	17	58	157	30	262
		Expected Count	17.8	64.9	154.5	24.8	262.0
		% within Job Involvement	60.7%	56.9%	64.6%	76.9%	63.6%
Total		Count	28	102	243	39	412
		Expected Count	28.0	102.0	243.0	39.0	412.0
		% within Job Involvement	100.0%	100.0%	100.0%	100.0%	100.0%

Figure 7: Chi- Square Test (Crosstabulation)

A closer look at the crosstabulation of gender and work involvement levels offers a better understanding of job involvement distribution between men and women in the surveyed population. Out of 150 females assessed for work involvement levels, 44 of them, representing 43.1% of the total, reported medium job involvement. In contrast, the sample included 262 more males than females. Males showcased higher job involvement in every category, with most demonstrating high or very high job involvement. 157 men, accounting for 64.0% of the sample, displayed high job involvement. Overall, this analysis sheds light on gender differences in work involvement and identifies potential areas for intervention within organizational contexts or further research.

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	5.197[a]	3	.158
Likelihood Ratio	5.370	3	.147
N of Valid Cases	412		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 10.19.

Figure 8: Chi- Square Test

The Chi-Square test statistic is 5.197 with 3 degrees of freedom, yielding an asymptotic significance of 0.158 (15.8%). Likewise, the Likelihood Ratio test statistic value is 5.370 with

identical degrees of freedom and a marginally lower asymptotic significance of 0.147 (14.7%). These numbers suggest a moderately low likelihood that the observed data would happen under the validity of the null hypothesis, though not low enough to discard the null hypothesis.

3.4. One Way Anova:

ANOVA

		Sum of Squares	df	Mean Square	F	Sig.
Environment Satisfaction	Between Groups	5.906	4	1.476	1.236	.295
	Within Groups	486.334	407	1.195		
	Total	492.240	411			
Job Involvement	Between Groups	4.023	4	1.006	1.907	.108
	Within Groups	214.606	407	.527		
	Total	218.629	411			
Job Satisfaction	Between Groups	3.718	4	.930	.768	.547
	Within Groups	492.699	407	1.211		
	Total	496.417	411			

Figure 9: One- Way ANOVA

In the analysis of Environment Satisfaction, Job Involvement, and Job Satisfaction, between-groups and within-groups comparisons are used to determine potential variations in mean scores across categories. For Environment Satisfaction, the F-statistic of 1.236 corresponds to a significance value (Sig.) of 0.295, suggesting a 15.8% probability of observing the data assuming the null hypothesis is true. Similarly, Job Involvement and Job Satisfaction analyses yield F-values of 1.907 and 0.768, respectively, with corresponding significance values of 0.108 and 0.547. In each case, since the significance values surpass the customary alpha level of 0.05, the null hypothesis cannot be dismissed. The within-groups comparisons further corroborate this outcome, as the observed data align with expectations under the null hypothesis.

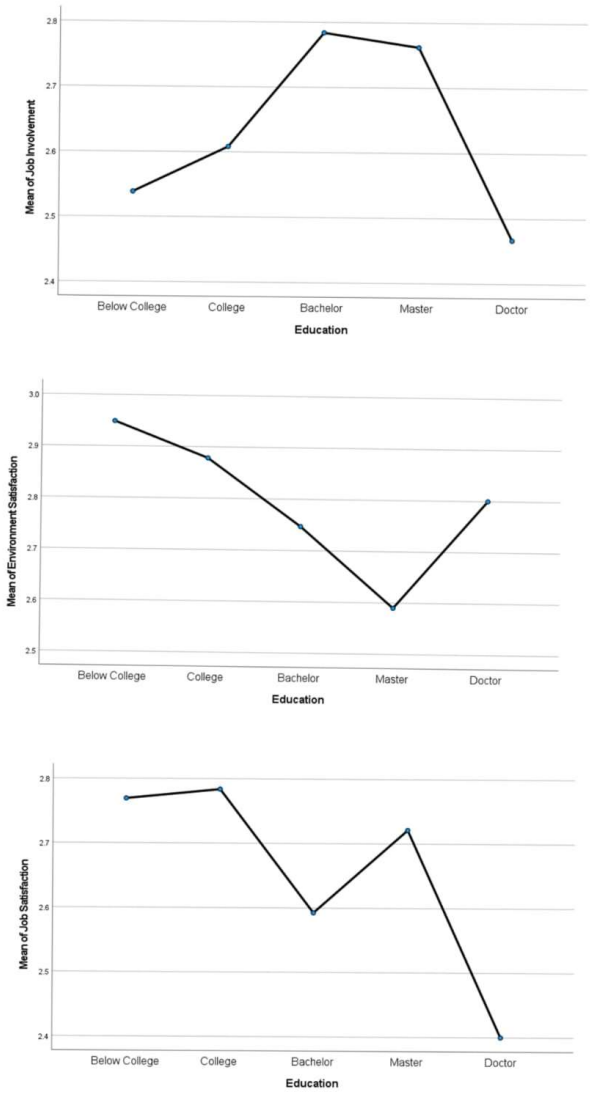

Figure 10: One- Way ANOVA (Variable Graphs)

In the Environment Satisfaction, Job Involvement, and Job Satisfaction analyses, F-statistics and significance values correspond to probabilities of 15.8%, 14.7%, and 54.7%, respectively, under the validity of the null hypothesis. As each significance value surpasses the customary alpha threshold of 0.05, the null hypothesis stands unrejected for all categories. This indicates an absence of significant discrepancies in group means across these dimensions, highlighting the lack of notable variations within the studied population.

3.5. Linear Regression Analysis:

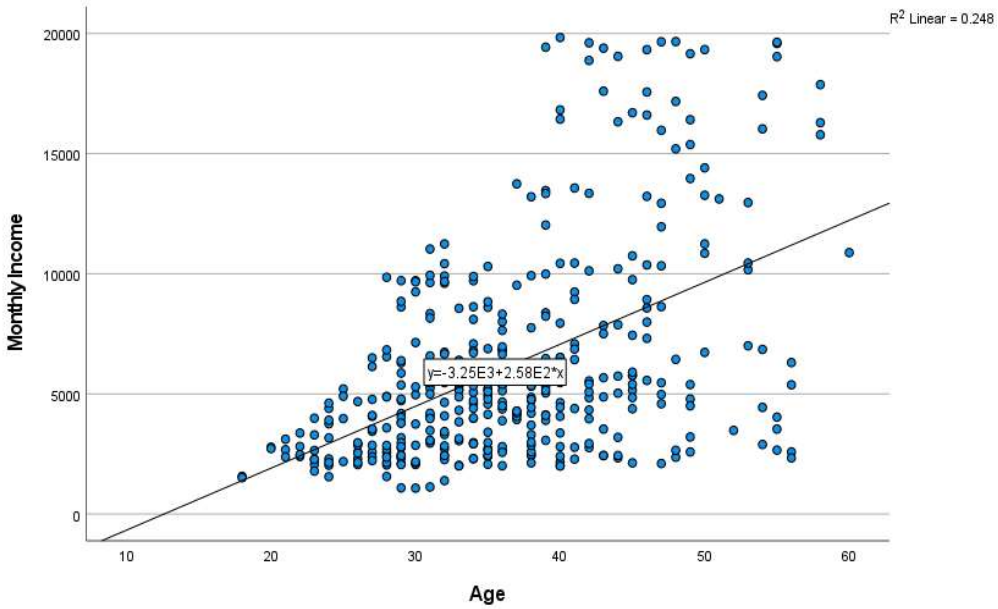

Figure 11: Scatter Plot of Linear Regression

A scatter plot graph illustrates the connection between age and monthly income, with individual age-income data points represented as blue dots. Ages from 10 to 60 are plotted along the horizontal axis, while monthly incomes from 0 to 20,000 are marked on the vertical axis. A trend line depicting a positive correlation indicates that monthly income tends to increase with age. Despite this, the trend line equation y = -3.25E+3 + 2.58E+2x and the low R^2 value of 0.248 reveal that age is not a major contributor to income variability. This suggests that factors other than age likely impact a considerable part of monthly salaries.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.498[a]	.248	.246	3774.887

a. Predictors: (Constant), Age

b. Dependent Variable: Monthly Income

Figure 12: Model Summary of Linear Regression

The model summary outlines the statistical association between age and monthly income, revealing a moderate positive correlation with an R value of 0.498. This suggests that as age rises, so does monthly income. However, the R Square value of 0.248 indicates that only about 24.8% of the variance in monthly income can be explained by age alone, leaving a considerable portion unaccounted for. The Adjusted R Square value of 0.246, slightly lower, adjusts for the number of predictors in the model, providing a more precise evaluation of its fit. The standard error of the estimate, measuring the average deviation of actual data points from the regression line, is 3774.887, showing a degree of variability surrounding predicted values. Overall, while age does affect monthly income to some extent, it is evident that other factors play a substantial role in determining earnings.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1923364642.5	1	1923364642.5	134.975	<.001[b]
	Residual	5842405951.9	410	14249770.614		
	Total	7765770594.5	411			

a. Dependent Variable: Monthly Income

b. Predictors: (Constant), Age

Figure 13: ANOVA in Linear Regression

The outcomes of the regression analysis reveal a significant association ($p < 0.001$) and an F-statistic of 134.975 among the variables Constant, Age, and Monthly Income. The sum of squares for the regression component, which has one degree of freedom, is about 1,923,364,642.5. The residual sum of squares, totaling around 5,842,405,951.9 with 410 degrees of freedom, results in a mean square of approximately 14,249,770.614. With 411 degrees of freedom, the monthly income's total sum of squares is approximately 7,765,770,594.5. Upon obtaining a p-value below 0.001 through regression analysis, the null hypothesis is rejected due to the demonstrated statistical significance. This leads to the rejection of the null hypothesis, which claims that Age and Constant do not affect Monthly Income.

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	95.0% Confidence Interval for B Lower Bound	95.0% Confidence Interval for B Upper Bound
1	(Constant)	-3246.812	833.036		-3.898	<.001	-4884.367	-1609.258
	Age	257.861	22.195	.498	11.618	<.001	214.231	301.492

a. Dependent Variable: Monthly Income

Figure 14: Coefficients (Linear Regression)

The regression model explores the relationship between age and monthly income, employing age as a substantial predictor of income fluctuation. With an intercept of around -$3246.812, the model indicates a theoretical baseline for monthly income when age is zero, although this is not applicable in real-world scenarios. An increase of roughly $257.861 in monthly income is expected for every additional year of age. The constant and age coefficients are vital in predicting income levels, as they are both statistically significant predictors ($p < 0.001$). Age exerts a moderate positive influence on monthly income, as indicated by its standardized coefficient (Beta) of approximately 0.498. Confidence intervals for the constant and age coefficients bolster the model's dependability, providing intervals within which the actual values are likely to reside. Collectively, this analysis accentuates age as a major factor affecting monthly income, thereby enhancing the understanding of income disparities within the dataset.

4. Reason of Proposed Hypothesis:

Statistical methods tailored to specific variables and research topics can be utilized to test the given hypotheses effectively. The chi-square test proves useful when examining relationships between categorical data such as demographic traits and employee satisfaction levels. In contrast, the one-way ANOVA test is suitable when comparing mean satisfaction levels across different groups like departments or leadership styles. Linear regression analysis can provide a deeper understanding of the relationship between independent factors, like work-life balance and leadership styles, and employee happiness. By applying these statistical techniques, researchers can draw informed conclusions about the hypotheses, identifying significant correlations and connections that enhance comprehension of employee satisfaction dynamics within organizations.

5. Result and Discussion:

The application of statistical techniques, such as the chi-square test, one-way ANOVA, and linear regression analysis, establishes a strong basis for assessing hypotheses in the domain of employee satisfaction research. These methods empower researchers to compare satisfaction levels across various groups, pinpoint significant correlations between categorical variables, and delve into the associations between independent factors and employee satisfaction. The strategic use of these statistical methods unveils crucial insights into the elements impacting employee satisfaction within organizations. This knowledge supports evidence-based decision-making and execution of targeted initiatives to improve workplace satisfaction and overall productivity.

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